

ECON 100: Introduction to Economics

PRACTICE Exam 1 SOLUTIONS, Fall 2026

- Define any variables and/or state any assumptions you need to answer the problems.
 - All materials should be put away before beginning the exam. Use of cell phones during the exam for any purpose is forbidden.
 - Please write your answers in the space provided.
 - Keep your answers short but clear. Your goal is to convince a skeptical grader that you understand the relevant concepts well enough to answer the question you are given.
 - The questions on the exam sum to 56 points.
 - Good luck on your Econ test!
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1. (2 points) What is your name?

Solution: Hopefully you got these 2 points.

2. (2 points) What section are you in?

Solution: Hopefully you got these 2 points as well.

1 True/False Questions

Indicate “T”rue or “F”alse for each of the following statements or claims. For each false statement, if you correctly and clearly explain why it is false, I will give you a bonus point. No explanations necessary for claims or statements. which you think are true.

3. (2 points) Bonomo’s (ice-cream shop down the hill) currently makes only ice cream cones. They decide to start making ice cream sundaes, but their equipment and workers are more suited/more experienced at making ice cream cones. **Claim:** Based on what we have learned in class, a Production Possibility Frontier (PPF) for Bonomo’s making ice cream cones and ice cream sundaes would be a straight line.

Solution: False. Bonomo’s inputs are not easily substitutable to produce the two goods, so you would expect increasing opportunity cost, and therefore the PPF would be bowed out.

4. (2 points) Suppose that Alexandria currently tutors economics at a rate of \$25 per hour, and is willing to tutor 10 hours per week. After one year, she decides to raise her rate to \$30 per hour, but now is only willing to tutor 8 hours per week. **Claim:** If we define wage elasticity of hours worked as how responsive hours worked are to hourly wage, Alexandria would have a positive such elasticity.

Solution: False. As wage increases, Alexandria’s hours worked decreases. Therefore this elasticity would be negative.

5. (2 points) Grace has devised a study plan for her 2 upcoming midterms in history and anthropology spread across 40 hours, which is the maximum she is willing to study for both classes. Both exams are graded 0-100. She then decides to add 1 extra hour to her study plan. She realizes with that extra hour, she expects she could increase her grade on her history exam by 1 point, and her grade on her anthropology exam by 1 point. **Claim:** Based on what we have learned in class, Grace’s study plan is consistent with utility maximizing behavior.

Solution: True.

6. (2 points) Joanna runs a car wash in Clinton, NY, and currently washes 10 cars per day at a price of \$10 each. Joanna decides to increase her price to \$12 per car wash. After the price change, her total revenue becomes \$96 dollars per day. **Claim:** Between these two prices, the absolute value of consumers' price elasticity of quantity demanded is greater than 1.

Solution: True.

7. (2 points) Jenny and Jared are two first graders playing on the playground. Jenny finds a cool rock, and Jared finds a cool leaf. After recess, their teacher notices Jenny with a leaf in her hair and Jared has a rock on their desk. **Claim:** Based on what we have learned in class, we cannot say anything for certain about how much Jared valued the leaf compared to the rock.

Solution: False. Jared agreed to trade the rock for the leaf, and would only agree to such a trade if they valued the rock just as much or more than the leaf. Therefore we can say for certain Jared valued the leaf at least as much as they valued the rock.

2 Multiple Choice Questions

Circle the best answer to each question. There is only one answer for each question. No explanation necessary.

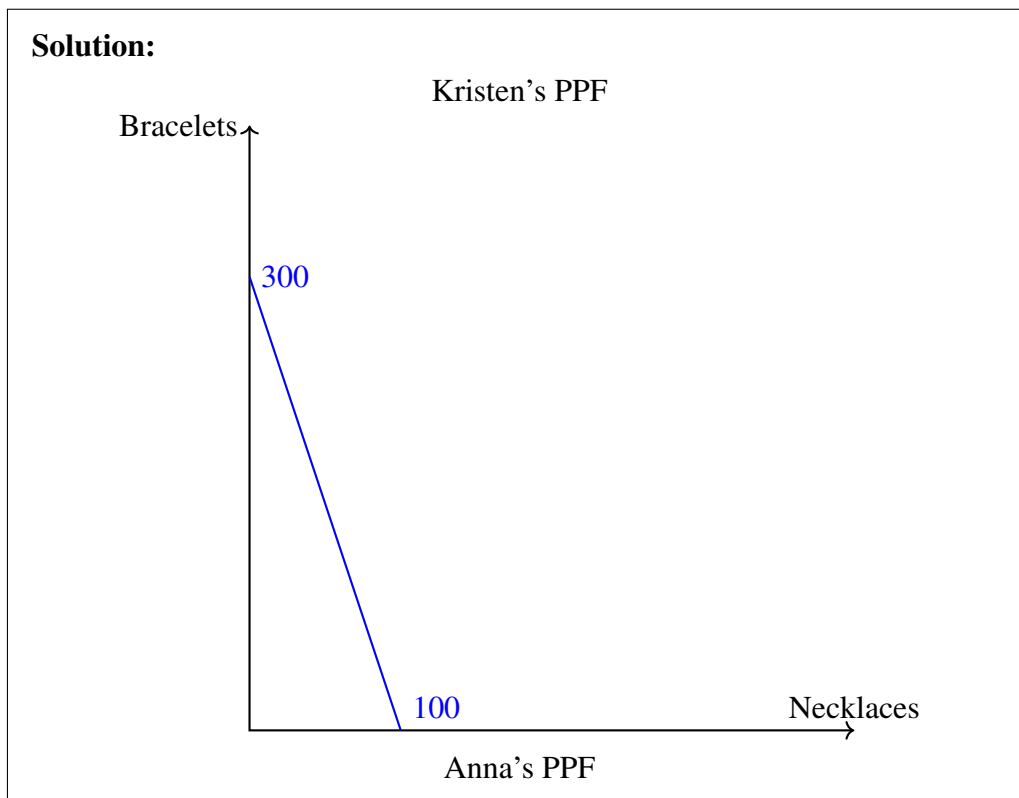
8. (2 points) You open up some data on trade between the United States and Colombia. You see that in 2025, Colombia imported 50 tons of soybeans from the United States, and the United States imported 100 tons of coffee from Colombia. **Question:** Assuming coffee and soybeans are the only two goods both countries produce, and based on what we have learned in class, which of the following **must** be true.
- A. The United States has the absolute advantage in producing soybeans.
 - B. Colombia has the absolute advantage in producing coffee.
 - C. The United States has a comparative advantage in producing coffee.
 - D. Colombia has a comparative advantage in producing coffee.**
9. (2 points) Suppose Sharon is selling Girl Scout selling cookies. Suppose there is only 1 type of Girl Scout Cookie. Sharon estimates that her marginal cost of selling Q boxes of Girl Scout cookies is given by $MarginalCost = 164 + 2Q$. Each box of Girl Scout cookies sells for \$4. **Question:** What is Sharon's profit maximizing quantity of Girl Scout cookie boxes she should sell?
- A. 80 boxes.**
 - B. 40 boxes.
 - C. 8 boxes.
 - D. 2 boxes.
10. (2 points) Suppose that Siobhan has set up a lemonade stand in the village of Clinton, and you decide to support their stand. Suppose your marginal benefit for Q cups of lemonade is given by $10 - Q$, and Siobhan's marginal cost to produce Q cups of lemonade is given by $6 + Q$. Which of the following represents a price (P) and quantity(Q) which would result in mutually beneficial trade?
- A. $Q=1, P=6$.
 - B. $Q=2, P=6$.
 - C. $Q=1, P=7$.**
 - D. $Q=2, P=7$.

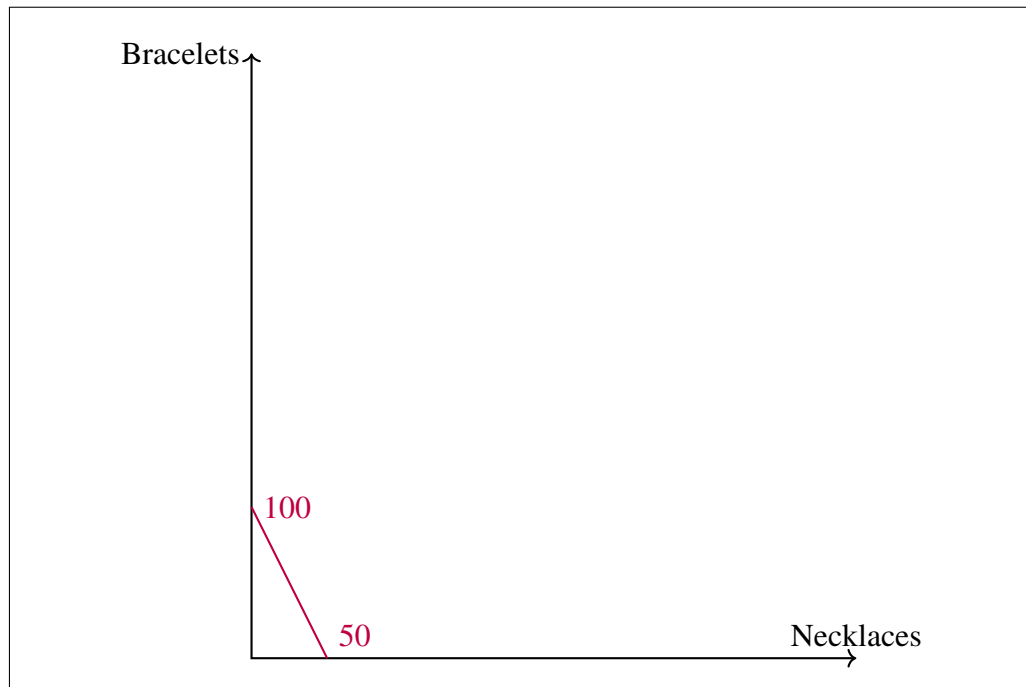
11. (2 points) Suppose quantity supplied is given by $Q = 6P$. Using the midpoint method, calculate the price elasticity of quantity supplied between when $P = 10$ and $P = 6$.
- A. 0.5
 - B. 1.
 - C. 1.5.**
 - D. 2.
12. (2 points) Suppose that when the price of a Ben and Jerry's pint is \$6 and the price of an oatmeal packet is \$2, Stephanie purchases 2 pints and 7 packets of oatmeal. When the price of a Ben and Jerry's pint drops to \$5, and the price of oatmeal is unchanged, Stephanie buys 1 pint and 8 packets of oatmeal. **Question:** Based only on the information above, what we have learned in class, and assuming Stephanie's purchases are consistent with utility maximizing behavior, which of the following is true for Stephanie?
- A. Ben and Jerry's pints are an ordinary good; pints and oatmeal are complements.
 - B. Ben and Jerry's pints are an ordinary good; pints and oatmeal are substitutes.**
 - C. Ben and Jerry's pints are a Giffen good; pints and oatmeal are complements.
 - D. Ben and Jerry's pints are a Giffen good; pints and oatmeal are substitutes.

3 Short Answer Questions

These questions all require an explanation. Remember you are trying to convince me you understand the why and the how of what you are doing, not simply getting the answer correct. Cite specific concepts from class in your answers for full credit.

13. **Etsy Craft Business.** Kristen and Anna own a small business in which they make bracelets and necklaces to sell on Etsy. Kristen can make 30 bracelets per hour or 10 necklaces per hour. Anna can make 10 bracelets per hour or 5 necklaces per hour. Kristen and Anna agree to spend no more than 10 hours per week making bracelets and necklaces to sell.
- (a) (8 points) Draw a separate Production Possibility Frontier (PPF) for each person, and be sure to label the axes. Also be sure to indicate which graph represents which person's PPF.





- (b) (4 points) Who has the absolute advantage in producing bracelets? Who has the absolute advantage in producing necklaces? Briefly explain your answer.

Solution: Kristen has the absolute advantage in both bracelets and necklaces, because in 10 hours she can make more bracelets (300 vs 100) and more necklaces (100 vs 50) than Anna.

- (c) (6 points) Suppose Anna and Kristen want to maximize the total number of bracelets and necklaces combined they produce each week. How should each of them divide their time between making bracelets and necklaces? Explain your answer.

Solution: Kristen's opportunity cost of 1 necklace is 3 bracelets, and Anna's opportunity cost of 1 necklace is 2 bracelets. Therefore Anna has the comparative advantage in necklaces, and Kristen has the comparative advantage in bracelets. Kristen will make 300 bracelets, and Anna will make 50 necklaces.

- (d) (6 points) Suppose each bracelet sells for \$1, and each necklace sells for \$2.50. If Anna and Kristen want to maximize their joint revenue, how should each of them divide their time between making bracelets and necklaces? Explain your answer.

Solution: An hour making bracelets for Kristen is worth \$30, while an hour making necklaces for Kristen is worth \$25. Her time always is better spent making bracelets, so she should make only bracelets (300 total for \$300). An hour making bracelets for Anna is worth \$10, while an hour making necklaces for Anna is worth \$12.50. Her time always is better spent making necklaces, so she should make only necklaces (50 total for \$125).

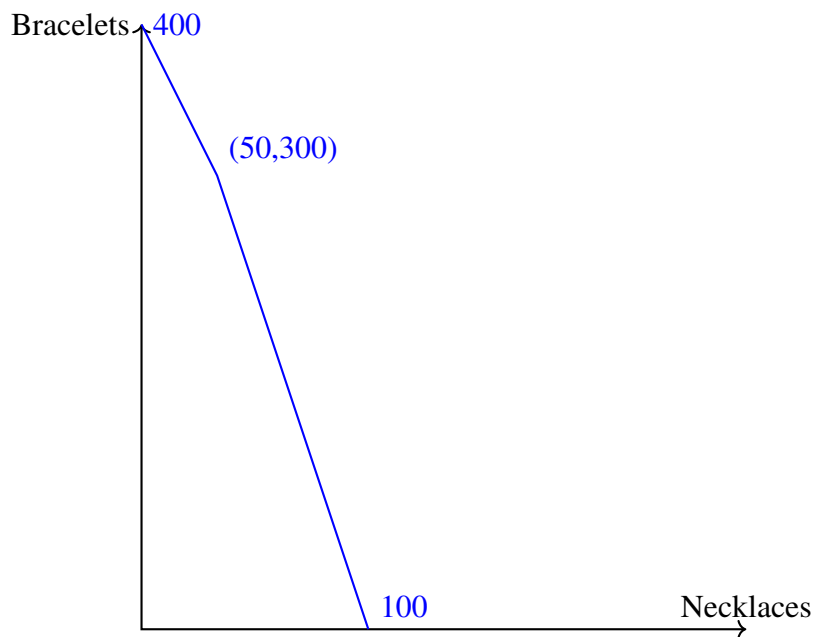
- (e) (2 points) What is Anna and Kristen's maximum joint revenue?

Solution: Their joint revenue will be \$425.

- (f) (6 points) Draw Anna and Kristen's combined PPF. Explain why the slope of the PPF is not constant even though both Anna and Kristen's individual PPFs are linear.

Solution:

Combined PPF



At (0,400) both Anna and Kristen are only making bracelets. From here, Anna switches to making necklaces first, she has the comparative advantage in producing necklaces. When she is only making necklaces and Kristen is only making bracelets, we get to the point (50,300). From there, Kristen starts switches to making bracelets at her opportunity cost of 3 bracelets:1 necklace, so the slope changes at (50,300).

Extra Scrap Page